

Proloy Karmakar

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🔗 ProloyKarmakar 📖 Google Scholar 📄 Portfolio Website 📊 R⁶ ResearchGate

EDUCATION

Bachelor of Science in Computer Science and Engineering

Khulna University of Science and Technology 🌐

01/2020 – 08/2025

Khulna, Bangladesh

CGPA: 3.86/4.00

Merit Position: 7th (Out of 121 students)

Higher Secondary School (HSC)

Notre Dame College, Dhaka 🌐

07/2017 – 04/2019

Dhaka, Bangladesh

GPA: 5.00/5.00

RESEARCH INTERESTS

- Machine Learning, Deep Learning, Natural Language Processing, Image Processing and Computer Vision, LLMs, Human Computer Interaction (HCI), Healthcare AI, Explainable AI (XAI)

PUBLICATION / RESEARCH EXPERIENCE

- Automated Pregnancy Risk Level Prediction Using Advanced Machine Learning and Deep Learning Algorithm
Authors: Proloy Karmakar, Md Sazzad Hossain, Riazul Islam, Mehedi Hasan
<https://www.ijcaonline.org/archives/volume186/number77/automated-pregnancy-risk-level-prediction-using-advanced-machine-learning-and-deep-learning-algorithm/> 🌐
- Early Detection of Suicidal Ideation from Multi-Platform Social Media Data: A Comparative Study of Traditional Machine Learning and Deep Learning Models
Authors: Proloy Karmakar, Chinmoy Modak Turjo
<https://ieeexplore.ieee.org/document/11172223> 🌐
- Human–AI Interaction in Healthcare: A User-Centered Evaluation of Trust and Usability in Explainable AI Interfaces
Authors: Proloy Karmakar (Corresponding Author), Shureed Sazzad
<https://doi.org/10.1109/QPAIN69676.2026.11546108> 🌐
- A Comprehensive GNN-Driven Approach for Detecting Automated Accounts and Analyzing Bot Coordination Patterns in Social Media Graphs (Accepted at the ICEFrontT: International Conference on Engineering and Frontier Technologies 2026)
Authors: Proloy Karmakar, Md. Abdul Hadi, Sabrin Alom, Nelay kumer Sagor, MD Sakib Al Hasan
- Exploring AI-driven solutions for breast cancer diagnosis through a comprehensive analysis of text and image modalities (Accepted at the IEEE CSBDC Student Symposium 2025)
Authors: Proloy Karmakar, Rakibul Hasan Adnan, Chinmoy Modak Turjo, Abtahe Alam

ONGOING RESEARCH

- Sentiment-Aware Embedding for Bangla Text: Enhancing Cross-Domain Sentiment Analysis with ELMo (Under Review)
Authors: Proloy Karmakar, Jerin Tasnim, Dr. K. M. Azharul Hasan
<https://s24.ieeecsbd.org/papers/55-cross-domain-sentiment-analysis-using-word-embedding> 🌐
- Modeling and Explaining Human Trust in AI Systems: A Multi-Factor Comparative Study in Human–AI Interaction (Under Review)
Authors: Proloy Karmakar, Farzana Akter, Suraiya Tabassum, Md. Moklachur Rahaman Safin
- Visualizing AI Uncertainty: An HCI Study on How Uncertainty-Aware Interfaces Influence Human Decision-Making (Under Review)
Authors: Proloy Karmakar (Corresponding author), Shureed Sazzad

PROJECTS

3D Interactive Chess Game

11/2024 – 02/2025

Developed a 3D interactive two-player chess game in OpenGL with realistic lighting, dynamic shading, and keyboard-controlled camera and piece movement.

Cartooning Image

07/2024 – 09/2024

Developed an image processing project in Python using OpenCV and NumPy, allowing users to upload images and apply cartooning effects like **Color Quantization**, **Comic Book Style**, and **Edge Detection**.

Checkers Game

07/2024 – 09/2024

Developed an AI-based Checkers game in Pygame using genetic algorithms, fuzzy logic, and minimax with alpha-beta pruning to enable intelligent and strategic gameplay.

Aalap- a chatting application

11/2023 – 04/2024

Created an Android application with Google login integration that enables users to chat, upload posts, and explore a dynamic news feed for interactive social engagement.

Furniture Shop App - an iOS applications

09/2023 – 12/2023

Developed an iOS furniture shop app using Swift, allowing users to log in, browse, search furniture, and make secure online payments for a smooth shopping experience.

Compiler Design Project

09/2023 – 12/2023

Developed a custom compiler using Lex and Yacc to handle directives, variable declarations, control structures, and functions with built-in error detection.

Digital Multimeter - A Hardware Project

06/2023 – 08/2023

Designed a digital multimeter that measures and displays current, voltage, and power consumption of a fan or light on an LCD, adjusting dynamically with fan speed.

Cardiac Recorder - A Software

05/2023 – 07/2023

Developed a blood pressure tracking app featuring secure authentication, OTP verification, health data visualization, history management, unit testing, and Javadoc documentation.

Hotel Management- A Database Project

04/2023 – 06/2023

Developed a "Hotel Management" database project using SQL and PL/SQL to efficiently manage bookings, customer data, and hotel services.

Hotel Management System - A website using Asp.net

01/2023 – 04/2023

Developed a hotel management system website using ASP.NET  , streamlining booking, management, and operations through a user-friendly interface.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java, Assembly
- **Data Analysis & Visualization:** Pytorch, Scikit-learn, TensorFlow, Matplotlib, Excel
- **Web and Mobile Development:** Android, Swift, ASP.NET  , HTML, CSS, JavaScript
- **Database Management:** MySQL, PL/SQL, SQLite
- **Operating Systems:** Windows, Linux
- **Version Control:** Git
- **Designing & Development Tools:** AutoCAD, LaTeX, OpenCV, OpenGL, Cisco Packet Tracer
- **Coursework & Topics Covered:**
 - Data Structures & Algorithms
 - Operating Systems
 - Computer Networking
 - Database Management System (DBMS)
 - Artificial Intelligence
 - Object-Oriented Programming
 - Web Development
 - Android App Development
 - Graphics Design using Modern OpenGL 3.3

PROFESSIONAL EXPERIENCE

Lecturer, Department of Computer Science and Engineering
Southeast University 

10/2025 – 06/2026
Dhaka, Bangladesh

VOLUNTEER ACTIVITIES AND COMMUNITY ENGAGEMENT

President of The United Club (TUC)

06/2016 – Present
Barishal, Bangladesh

Founder of The United Club (TUC)

A social development non-profit voluntary organization

06/2016 – Present
Barishal, Bangladesh

Executive Member at BD Clean

04/2020 – 04/2021
Barishal, Bangladesh

Member at Dream-Voluntary Blood Donation Society of KUET

01/2020 – 09/2025
Khulna, Bangladesh

Executive Member at Organization of KUET Sports (OKS)

01/2020 – 09/2025
Khulna, Bangladesh

SCHOLARSHIPS AND AWARDS

Deans's Award

4 consecutive awards

The Dean's Award is given to students having a CGPA of 3.75 or above in an academic year.

Education Board Scholarships

Barisal Board

- General grade scholarship in the SSC exam given by the Board of Intermediate & Secondary Education, Barisal

Online Course Certificate

- "Basics of Python" by Uniathena
- "Art of Problem Definition" by Passport to Earning Bangladesh.
- Intra Kuet Programming Contest Certificate by the Department of CSE, KUET.

Conference Certificates

- Paper Presentation certificate at CS BDC SYMPOSIUM 2024 at Jagannath University
<https://s24.ieeecsbdc.org/papers/55-cross-domain-sentiment-analysis-using-word-embedding> 
- Paper Presentation certificate at CS BDC SYMPOSIUM 2025 at HSTU

Internship

- Encryptix, IT Services, and IT Consulting, New Delhi, Delhi, India